

- By Theorem 4 from Chapter 1, there is a pivot position in every row.
- By defn of pivot, it corresponds to the leading 1 of a row in rref form of the matrix.
- By the Row Reduction Algorithm and Theorem 1, matrix  $A$  is converted to a unique rref.
- By the defn. of rref, from Condition 4, we know that each leading entry is a 1.
- From Condition 2, we know that for each ~~column~~<sup>row</sup> having a leading entry, all the rows above it must have their leading entries in a column to the left of the given column.
- $\therefore$  The no. of columns of  $A$  must be <sup>at least</sup> as large as the no. of rows of  $A$ .

□

25) Suppose  $A$  is an  $m \times n$  matrix and there exist  $n \times m$  matrices  $C$  and  $D$  such that  $CA = I_n$  and  $AD = I_m$ . Prove that  $m = n$  and  $C = D$ .  
[Hint: Think about the product  $CAD$ .] (Fantastic problem)

**Solution:**

$$CAD = (CA) \cdot D = I_n \cdot D \text{ (Given)} = D \text{ (Theorem 2(e))}$$

$$(CA) \cdot D = C \cdot (AD) \text{ (Theorem 2(a))}$$

$$= C \cdot I_m \text{ (Given)}$$

$$= C \text{ (Theorem 2(e))}$$

$$\therefore C = D$$

By ~~Prob~~<sup>Prob</sup> 23, we know  $m \leq n$ . By Prob 24, we know  $m \geq n$ .

$$\therefore m = n \quad \square$$

In Exercises 27 and 28, view vectors in  $\mathbb{R}^n$  as  $n \times 1$  matrices. For  $u$  and  $v$  in  $\mathbb{R}^n$ , the matrix product  $u^T v$  is a  $1 \times 1$  matrix, called the **scalar product**, or **inner product**, of  $u$  and  $v$ . It is usually written as a single real number without brackets. The matrix product  $uv^T$  is an  $n \times n$  matrix, called the **outer product** of  $u$  and  $v$ . The products  $u^T v$  and  $uv^T$  will appear later in the text.

27. Let  $u = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix}$  and  $v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ . Compute  $u^T v$ ,  $v^T u$ ,  $uv^T$  and  $vu^T$ .